

LoRA-MoE Crowdsourcing Instruction Generation

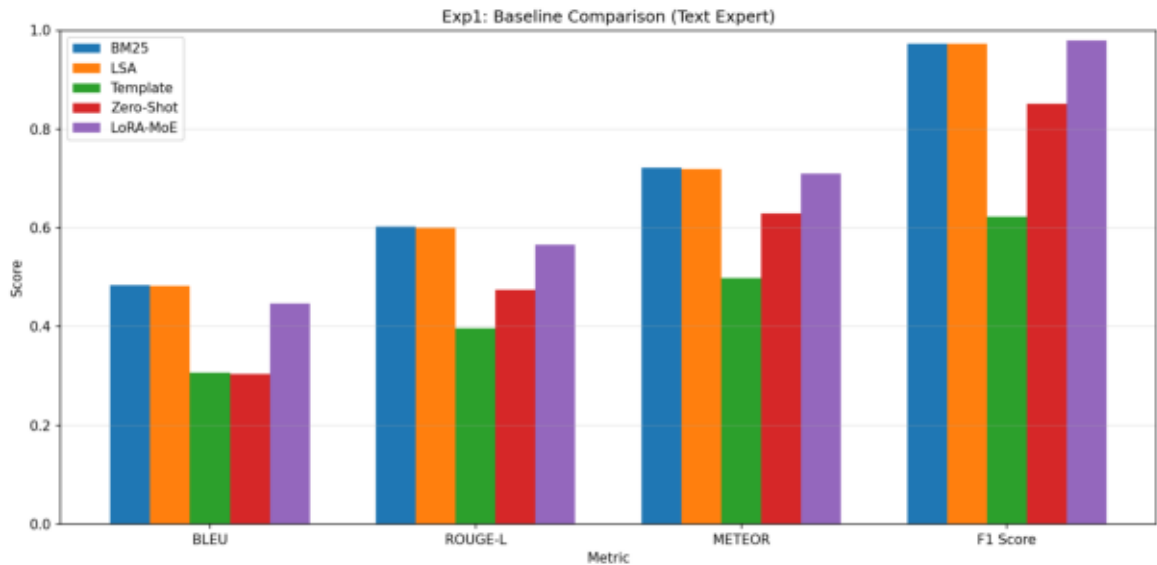
Comprehensive Experiment Report

Generated: 2026-02-28 17:09:11

Experiments included: Exp1, Exp2, Exp3, Exp4, Exp5, Exp6, Exp7
Base model: Qwen3-8B | Fine-tuning: LoRA-MoE (4 experts)
Expert types: Text, Image, UML, General

Key Results: LoRA-MoE Performance			
Text Expert:	ROUGE-L=0.5656	F1=0.9794	BLEU=0.4464
Image Expert:	ROUGE-L=0.7354	F1=1.0000	BLEU=0.6171

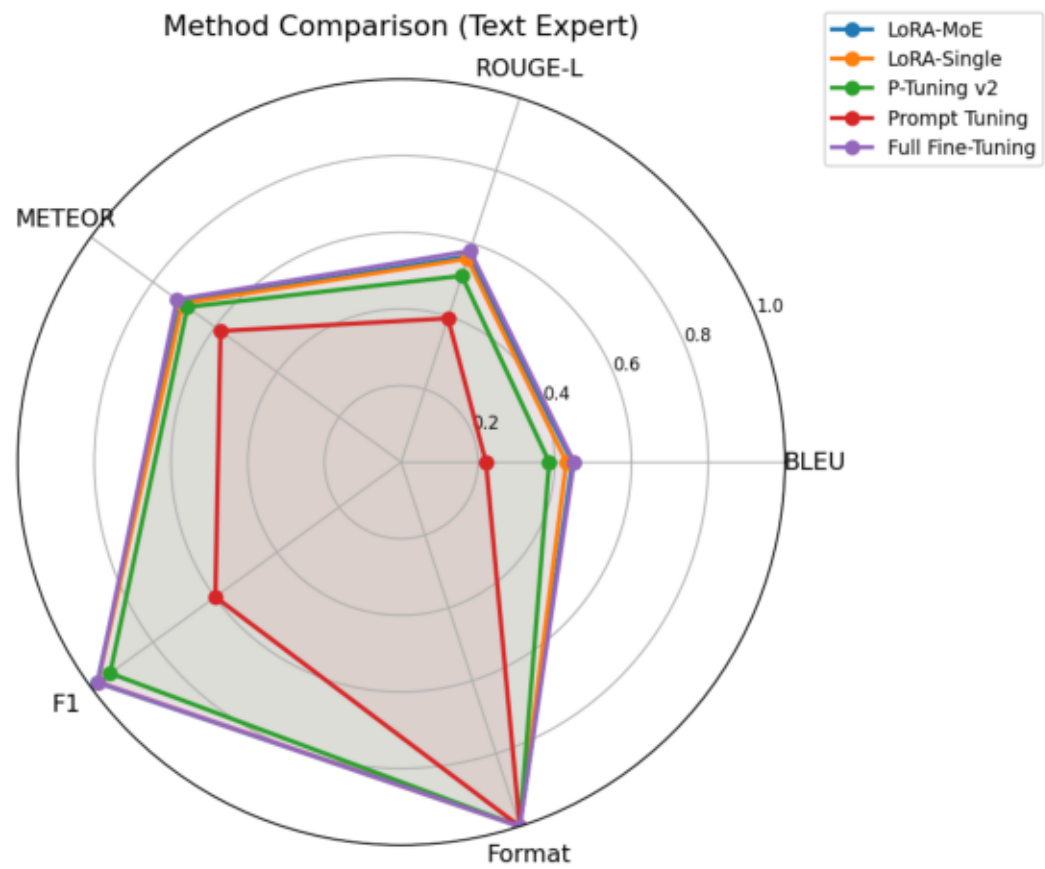
Experiment 1: Baseline Comparison (Text Expert)



Method	ROUGE-L	F1
BM25	0.603	0.973
LSA	0.601	0.973
LoRA-MoE	0.566	0.979
Zero-Shot	0.474	0.852
Template Fill	0.397	0.622

Observation: IR-based retrieval (BM25/LSA) achieves high ROUGE-L by copying the most similar instruction from the training set, but LoRA-MoE achieves the highest F1 score (0.979), indicating superior format compliance and structural quality. Zero-shot and Template methods lag significantly, confirming the value of fine-tuning over naive approaches.

Experiment 2: Fine-Tuning Method Comparison

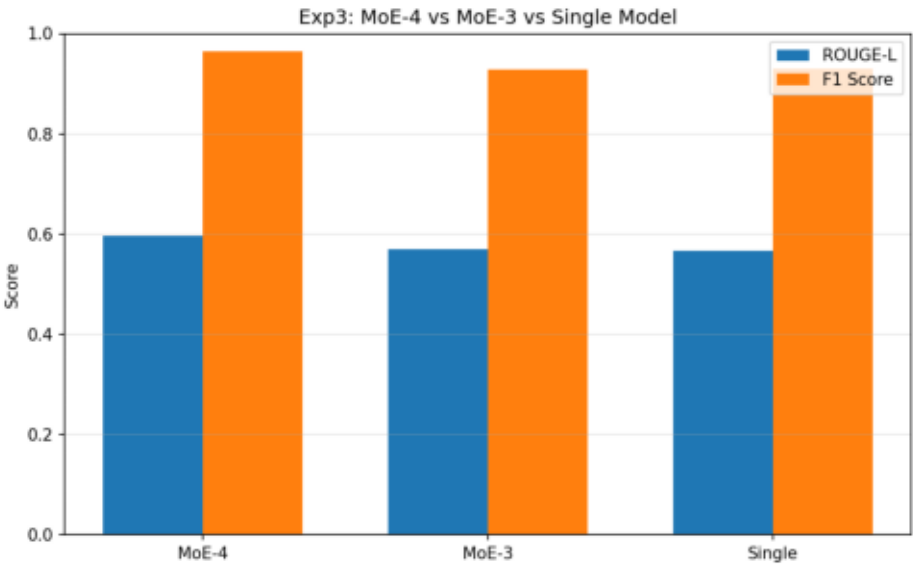
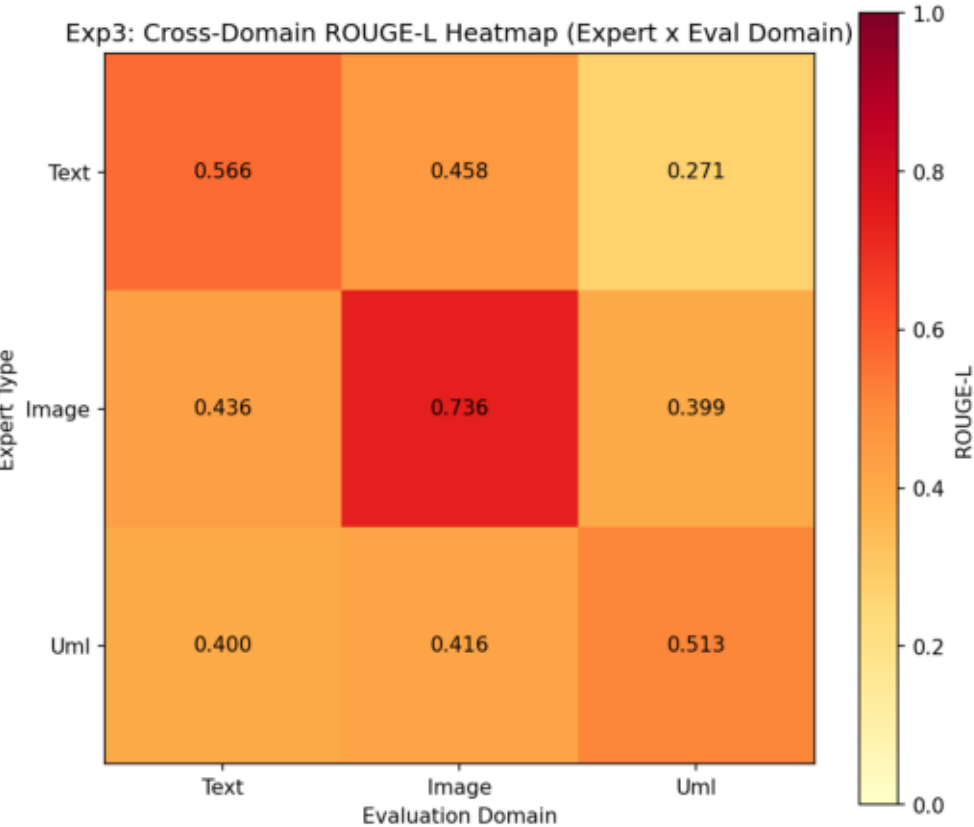


Text Expert Results:

Method	ROUGE-L	F1
LoRA-MoE	0.5656	0.9794
LoRA-Single	0.5582	0.9773
P-Tuning v2	0.5115	0.9379
Prompt Tuning	0.3947	0.5989
Full Fine-Tuning	0.5797	0.9773

Observation: LoRA-MoE achieves competitive performance with parameter-efficient LoRA adapters. Full Fine-Tuning achieves the highest ROUGE-L but requires significantly more resources. P-Tuning v2 performs reasonably while Prompt Tuning underperforms, suggesting that LoRA-based methods strike the best balance between quality and efficiency.

Experiment 3: MoE Architecture Validation

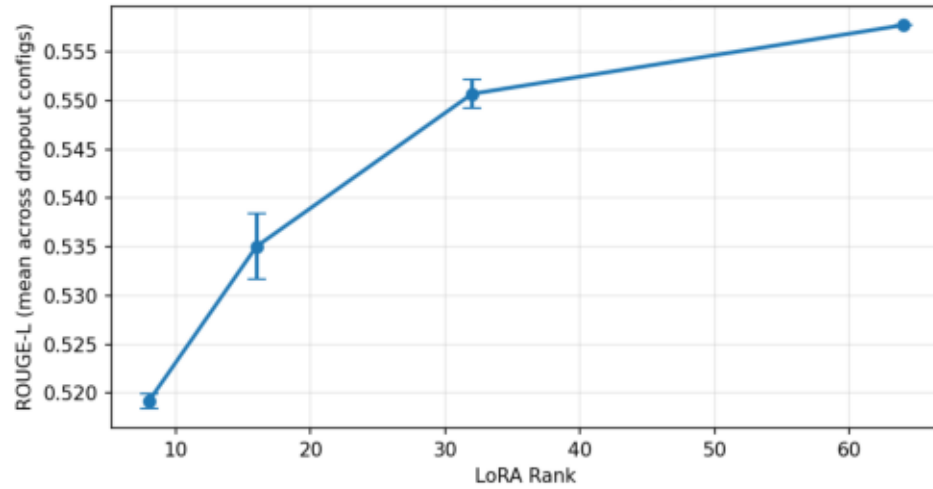


Architecture Results: MoE-4: ROUGE-L=0.5966 F1=0.9645 | MoE-3: ROUGE-L=0.5706 F1=0.9277 | Single: ROUGE-L=0.5664

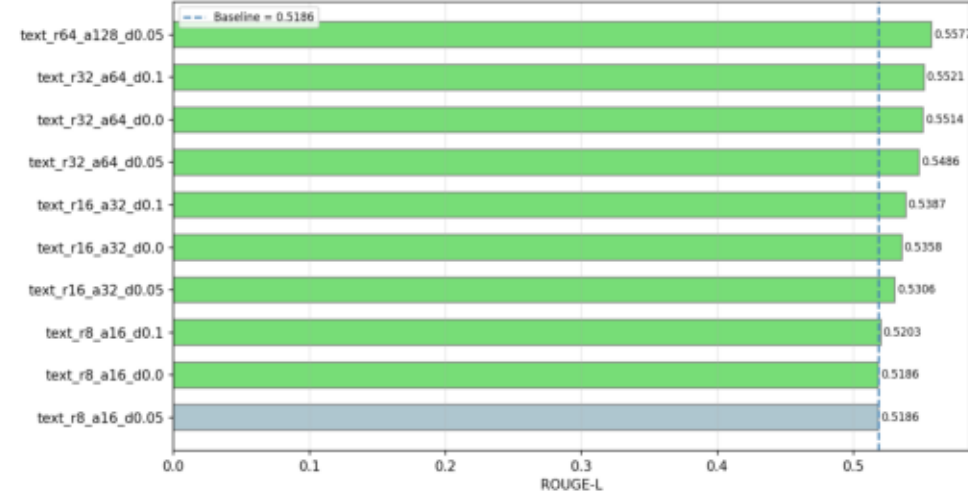
F1=0.9300 Cross-domain heatmap shows ROUGE-L when expert i is evaluated on domain j. Diagonal = matched expert (expected best). Off-diagonal = cross-domain penalty demonstrates specialization value of MoE routing.

Experiment 4: LoRA Hyperparameter Optimization (Text Expert)

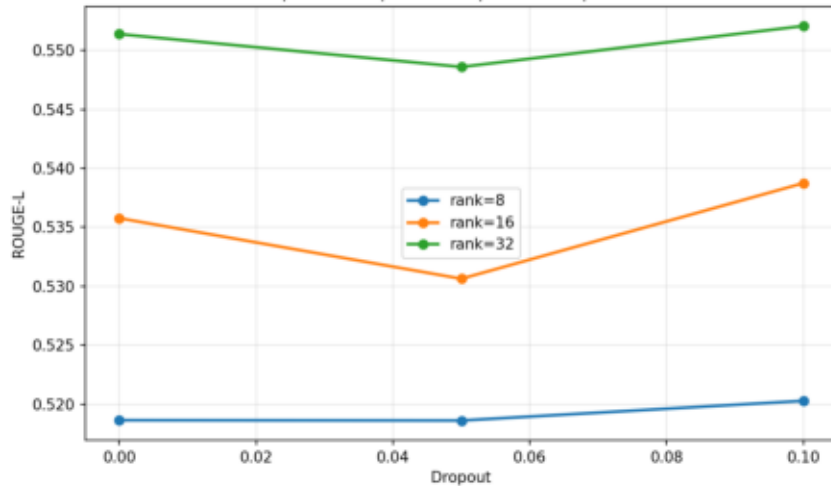
LoRA Rank Optimization (Exp4)



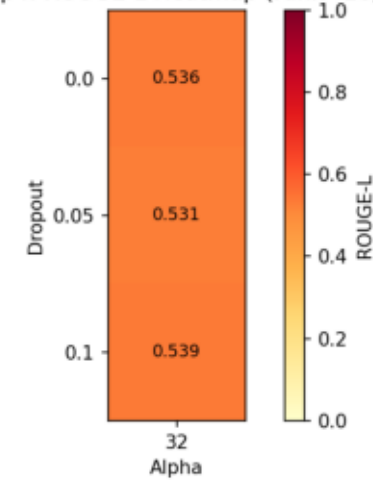
Exp4: Text Expert — All LoRA Configs (green > baseline, red < baseline)



Exp4: Text Expert — Dropout Effect per Rank

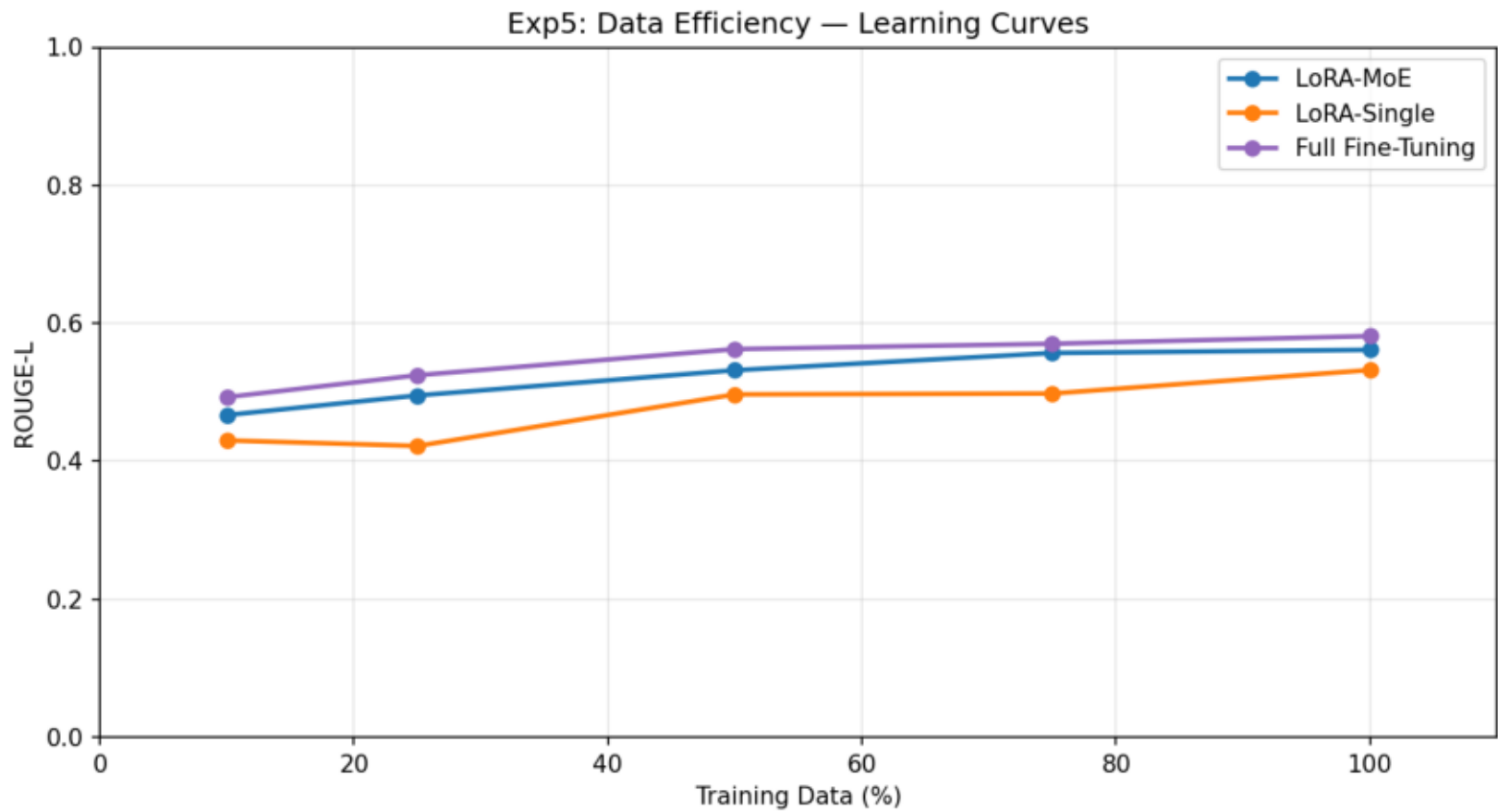


Exp4: ROUGE-L Heatmap (rank=16)



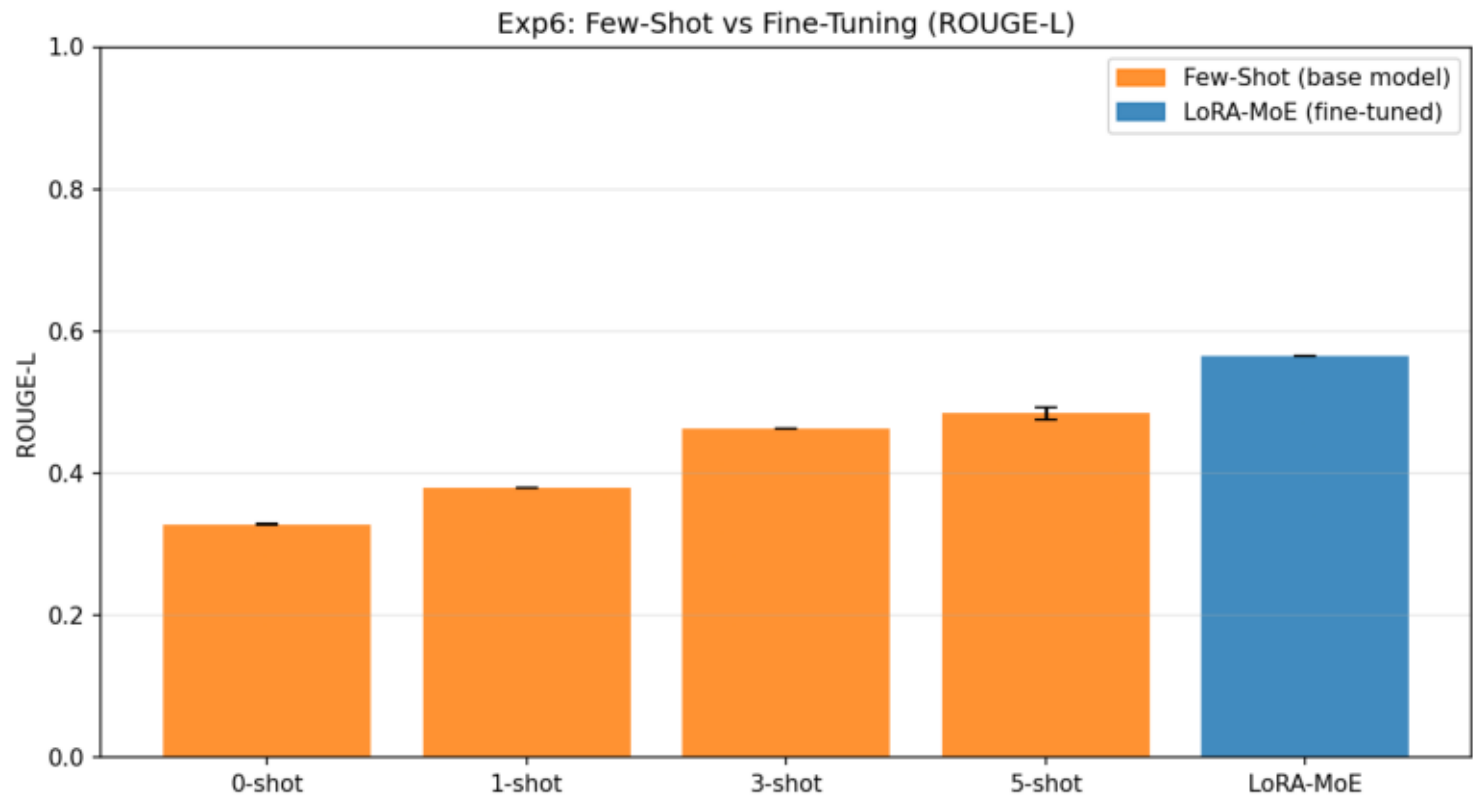
Best Config: text_r64_a128_d0.05 (ROUGE-L=0.5577) | rank=64, alpha=128, dropout=0.05

Experiment 5: Data Efficiency Analysis



Data efficiency analysis demonstrates how performance scales with training data fraction (10%~100%) across LoRA-MoE, LoRA-Single, and Full Fine-Tuning. LoRA-MoE is expected to be most data-efficient due to expert specialization, achieving strong performance even with limited training samples.

Experiment 6: Few-Shot vs Fine-Tuning | Summary



LoRA-MoE per-expert ROUGE-L summary:

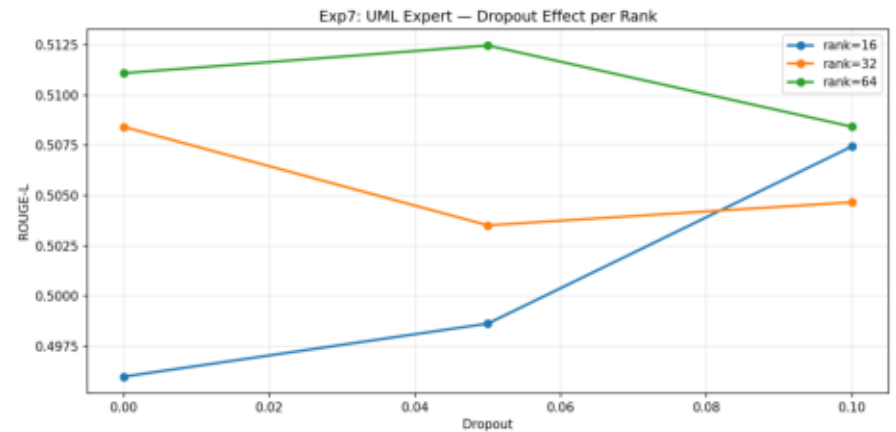
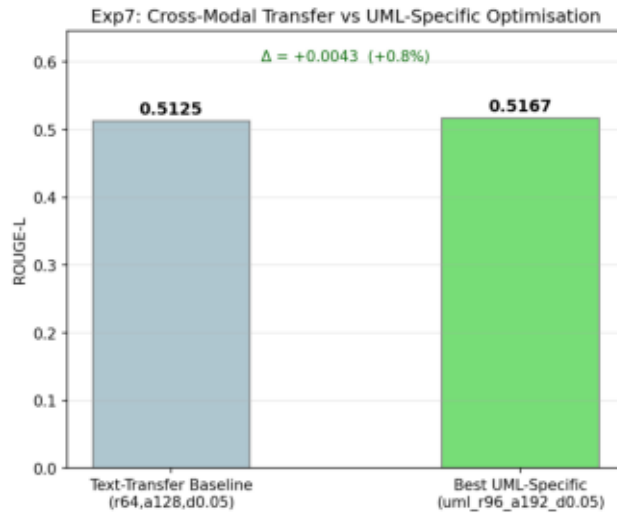
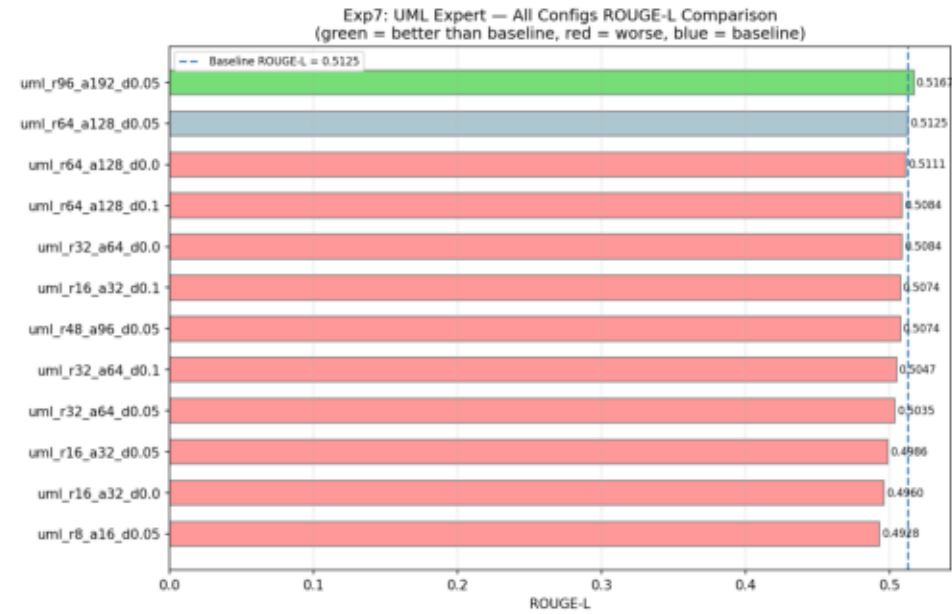
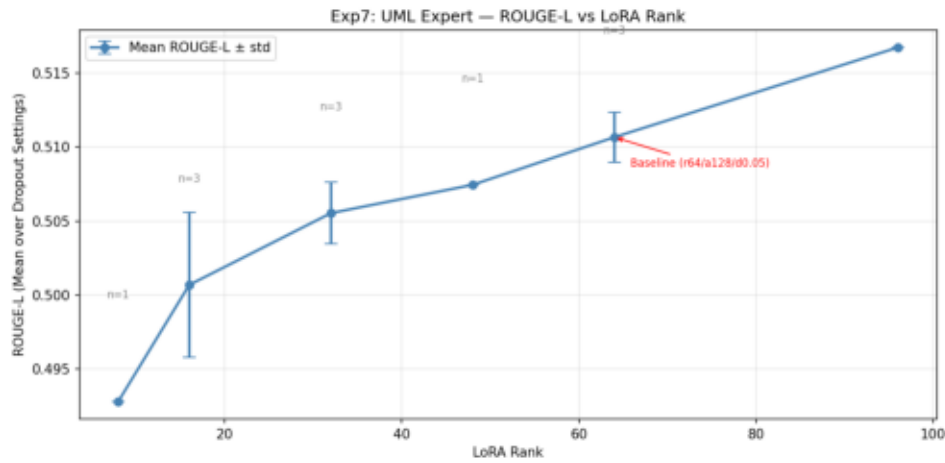
Text: 0.5656
Image: 0.7354
UML: 0.5151
General: 0.5515

Few-shot gap: 5-shot=0.4844 vs LoRA-MoE=0.5654

Exp4 best (Text): text_r64_a128_d0.05 ROUGE-L=0.5577

Exp7 best (UML): uml_r96_a192_d0.05 ROUGE-L=0.5167

Experiment 7: UML Expert LoRA Hyperparameter Optimization



Best UML Config: um_l_r96_a192_d0.05 (ROUGE-L=0.5167) | Δ vs text-transfer baseline: +0.0043